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Development of an Automated Screening System for Retinopathy of Prematurity Using a Deep Neural Network for Wide-Angle Retinal Images

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ABSTRACT Retinopathy of prematurity (ROP) is one of the main causes of childhood blindness. However, insufficient ophthalmologists are qualified for ROP screening. The objective of this paper is to evaluate the performance of a deep neural network (DNN) for the automated screening of ROP. The training and test sets came from 420 365 wide-angle retina images from ROP screening. A transfer learning scheme was designed to train the DNN classifier. First, a pre-processing classifier separated unqualified images. Then, pediatric ophthalmologists labeled each image as either ROP or negative. The labeled training set (8090 positive images and 9711 negative ones) was used to fine-tune three candidate DNN classifiers (AlexNet, VGG-16, and GoogLeNet) with the transfer learning approach. The resultant classifiers were evaluated on a test dataset of 1742 samples and compared with five independent pediatric retinal ophthalmologists. The receiver operating characteristic (ROC) curve, ROC area under the curve, and precision–recall (P-R) curve on the test dataset were analyzed. Accuracy, precision, sensitivity (recall), specificity, F1 score, the Youden index, and the Matthews correlation coefficient were evaluated at different sensitivity cutoffs. The data from the five pediatric ophthalmologists were plotted in the ROC and P-R curves to visualize their performances. VGG-16 achieved the best performance. At the cutoff point that maximized F1 score in the P-R curve, the final DNN model achieved 98.8% accuracy, 94.1% sensitivity, 99.3% specificity, and 93.0% precision. This was comparable to the pediatric ophthalmologists (98.8% accuracy, 93.5% sensitivity, 99.5% specificity, and 96.7% precision). In the screening of ROP using the evaluation of wide-angle retinal images, DNNs had high accuracy, sensitivity, specificity, and precision, comparable to that of pediatric ophthalmologists.

INDEX TERMS Retinopathy of prematurity, deep neural network, transfer learning, wide-angle retinal image, image classification, computer-aided diagnosis.

I. INTRODUCTION

Retinopathy of prematurity (ROP) is a vascular proliferative retinal disease affecting premature and low birth-weight infants. The incidence of ROP in infants whose birth weight $\leq$ 1500g in China is 26.0% [1]. It is one of the main causes of blindness in premature infants, with 400-600 cases each year in the United States [2], [3] and several thousand potential cases in China (see supplement). In addition, ROP leads to strabismus in 30% of the infants by 9 months of age [4].

Early screening and timely intervention are among the key factors in preventing blindness due to ROP and the treatment window for ROP is small [5]. In last decades, the incidence of ROP and severe ROP increased steadily because of the improvements in the survival of premature infants and unbalanced medical resources among different geographical regions. There are many obstacles in ROP screening in developing countries [6]. First, there is a lack of medical equipment and staff for ROP screening. In addition, the training of the

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ophthalmologists is not standardized and few ophthalmologists are qualified for ROP screening. Furthermore, there is an inadequate implementation of ROP screening policy in developing countries. Therefore, many premature infants become blind due to lack of timely screening and early treatment in underdeveloped regions. In China alone, with the new two-child policy and the increase in the number of older pregnant women, an increasing number of premature and very low birth weight preterm babies are emerging [7].

In the past half century, related work on computer-aided diagnosis (CAD) can be generally classified into two methodologies: ''rule-based'' and ''data-based''. Rule-based CAD is the traditional way of implementing artificial intelligence (AI) in medicine. It requires formally representing domain knowledge in computerized forms, such as production rules or frames. Some of the early rule-based CAD systems include MYCIN [8], INTERNIST [9], Isabel [10], and some systems for diabetic retinopathy (DR) [11]. The emergence of deep neural network (DNN) is a recent milestone of data-based CAD and it has been successfully used in several highly complicated medical domains including skin cancer classification [12], early diagnosis of Alzheimer's disease [13], syndrome diagnosis of chronic gastritis [14], and retinal vessel detection in fundus images [15], among others. DNN-based AI systems demonstrate several advantages compared to their human counterparts: 1) high speed and efficiency; 2) free of fatigue; 3) stable performance and unaffected by emotions; 4) fast self-learning and performs better with growing data; and 5) configurable, e.g. can increase recall levels at the cost of losing precision. In the specific domain of ROP, traditional research focuses on using traditional medical image processing and feature engineering techniques to detect pathological structures such as blood vessels and fundus lesions [16]. In December 2016, researchers from Google began to use a DNN to diagnose DR [17] and the DNN outperformed the median of ophthalmologists in both sensitivity and specificity. DNN has also been applied for the diagnosis of cardiovascular risk factors from fundus photographs [18]. This implies the feasibility of using DNN in ROP, as DR and ROP are both fundus-related diseases. Nevertheless, the two diseases have different etiologies, pathophysiologies, and manifestations, and the viewport and quality of wide-angle retinal images used in ROP screening are also different from DR fundus photographs. In May 2015, i-ROP, an AI that uses DNN for the diagnosis of ROP pre-plus and plus diseases, was launched [19], [20]. Worrall *et al*. [21] reported that it was possible to train a DNN for the diagnosis of ROP without explicitly specifying discriminative rules. Brown *et al*. [22] published their results on a novel DNN that can diagnose ROP plus disease with an accuracy at least as good as human experts'. Nevertheless, some limitations remain, such as the detection of the key pathological structures in the retina such as demarcation line, ridge and detachment.

Since it has been shown that telemedicine can be used for the diagnosis of ROP [23]–[25] and to address the above issue, we initiated the Telemed-R (Telemedicine system for ROP) project in June 2013. This system covers 30 hospitals (refer to RCG member list in supplement) in Guangdong and Fujian Provinces of China, and it was designed to support cross-regional telemedicine for ROP screening. In the beginning, Telemed-R relies on pediatric ophthalmologists to read the wide-angle retinal images uploaded by various partner hospitals. To alleviate pediatric ophthalmologists of such laborious and repetitive task, and also improve the stability of diagnosis quality, this study aimed to develop a computer-aided screening algorithm for ROP since 2016, which is a key module in Telemed-R.

II. METHODS

A. CLINICAL SETTINGS

Shenzhen Eye Hospital (SEH) is a 200-bed Class III specialized eye hospital in China. SEH has long been providing ROP-screening services for peripheral partner hospitals, as the leader of the ROP Collaboration Group (RCG). The RCG comprises of 30 hospitals (refer to supplement). All RCG members use the same ROP screening criteria, which was published by Ministry of Health of PRC in 2004. Preterm infants who receive ROP screening should meet one of the following criteria: 1) birth weight <2,000 g and 2) preterm infants with birth weight 2,000 g but having severe systemic disorders (according to pediatricians' assessment). With more than 10 years of experience, SEH has accumulated screening data from more than 20,000 premature infants. The RCG institutional review board agrees that approval and registration are not required for this study.

B. INFORMATION SYSTEMS

The architecture of Telemed-R is shown in Fig. 1. The first version of Telemed-R was built in 2013. It now contains several sub-systems. 1) The CMS-R (Case Management System for ROP), a web-based data management system which can be accessed by technicians and pediatric ophthalmologists from various areas. The technicians can input patient-related data and upload wide-angle retinal images to the central data repository of the CMS-R. As in July 2018, CMS-R has collected data from 26,424 infants. All these infants underwent ROP screenings by senior pediatric ophthalmologists in SEH. The demo version of the CMS-R can be accessed at http://ropd.brahma.top. 2) The Label-R (Image Labeling system for ROP) allows pediatric ophthalmologists to label wide-angle retinal images in the central data repository in order to provide training data set for the next machine learning process. Each image can be labeled with a pre-coded diagnostic terminology, and the physiological and pathological regions in each image can be annotated with rectangles, circles or polygons. Label-R can be accessed at http://label.brahma.top, and details are provided in supplement. 3) The CAD-R (Computer-Aided Diagnosis system for ROP) includes a ROP screening algorithm based on

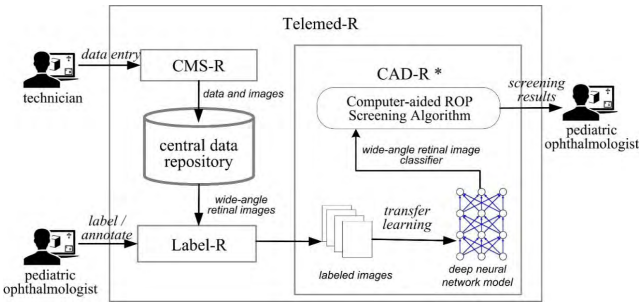


FIGURE 1. Telemed-R system architecture. Telemed-R: Telemedicine system for ROP; CMS-R: Case Management System for ROP; Label-R: image Labeling system for ROP; CAD-R: Computer Aided Diagnosis System for ROP.

DNN models which are trained by the labeled data set from Label-R.

C. SUBJECT DEMOGRAPHICS AND DATA SETS

As in July 2018, Telemed-R screening data contained 26,424 infants and 420,365 related images. These images are from the screening procedures carried out by the RCG. These images were taken using Retcam2 or Retcam3 camera by a professional technician with 10 years of experience. During each screening, each eye of premature infants was photographed in ten standard positions. The training set and test set used in this study are disjoint subsets of these images.

D. PEDIATRIC OPHTHALMOLOGISTS

Five pediatric ophthalmologists participated in this study. Two were senior experts (chief physicians), two were attending physicians, and one was resident. The two chief physicians have practiced ROP screening and treatment for more than 20 years; the two attending physicians, more than 10 years; the resident, 3 years. They screened about 2,000 cases and treated about 50 cases annually. All the pediatric ophthalmologists independently labeled the images as positive or negative. In case of discrepancies, consensus via group discussion was used as the final result.

E. TRANSFER LEARNING FRAMEWORK FOR ROP SCREENING

General-purposed DNNs such as AlexNet, VGG, and GoogLeNet (trained on ImageNet) are able to extract general visual features and advanced semantics from images. The basic idea behind transfer learning is that these automatically extracted features by the pre-trained models can be reused to serve other classification and detection tasks in different domains [26].

The designed transfer learning workflow (Fig. 2) is as follows. 1) Training of a preprocessing classifier to remove unqualified images. These images include highly-blurred images (very common for newborns), very dark or bright images, and non-fundus photographs. The preprocessing classifier is fine-tuned based on the VGG-16 DNN model. 2) Application of the binary classifier to preprocess the

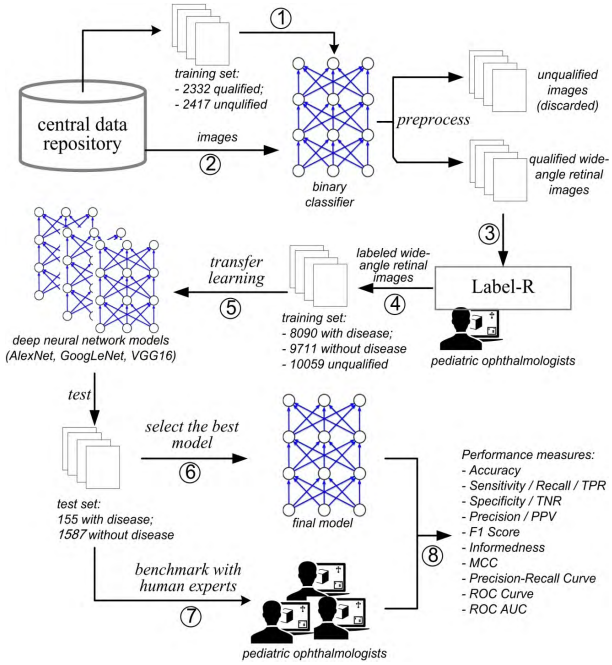


FIGURE 2. The transfer learning framework for ROP (retinopathy of prematurity) screening. (1) Training of a preprocessing binary classifier to recognize unqualified images, including highly-blurred images, very dark or bright images, and non-fundus photographs. (2) Application of the binary classifier to preprocess the entire image set. (3) Importation of the qualified images to the Label-R for labeling. (4) The pediatric ophthalmologists label and validate the images to construct a training set. (5) Training of the DNNs by the transfer learning approach. (6) Evaluation of the candidate models' performances on the test set. (7) Evaluation of the performances of human pediatric ophthalmologists on the same test set. (8) Performance comparison.

entire image set. 3) After preprocessing, 37,443 unqualified images were excluded from the original 420,365 images. The remaining images were then imported to the Label-R and randomly shuffled to split into five equal sets; each set was assigned to a pediatric ophthalmologist. 4) As the number of normal wide-angle retinal images are nearly ten times the number of ROP ones, we randomly picked similar quantities of both classes from the first-round labeled image set in order to make a balanced training set. The selected images were then cross-validated by senior pediatric ophthalmologists to ensure labeling quality. The final training set contained 8090 wide-angle retinal images with ROP and 9711 without ROP. 5) The DNNs were trained by the transfer learning approach. Three pre-trained DNNs were used in this study: AlexNet (8 layers), VGG-16 (16 layers), and GoogLeNet (22 layers). All of them are gold or silver winners of the ILSVRC (ImageNet Large Scale Visual Recognition Challenge) contests. The fine-tuning details for the three models are listed in Table 1. After training, the three models' classification accuracies on the validation set were 91.1%, 97.9%, and 98.1%, respectively. 6) The models' performance was evaluated on the test set. The test set was randomly selected. It was composed of 155 wide-angle retinal images with ROP and 1587 without ROP. The ratio of the two classes

TABLE 1. Deep neural network fine-tuning parameters.

Model Type	AlexNet	VGG-16	GoogLeNet
Layers	8	16	22
ImageNet top-5 error	0.164	0.066	0.067
Retrained layers	fc8	fc8	loss1, 2, 3/ classifier
Learning rate ratio (retrained layer: others)	5:1	5:1	5:1
Validation accuracy after training	0.911	0.979	0.981

DNN = deep neural network. ILSVRC = ImageNet Large Scale Visual Recognition Challenge. The VGG-16 used in this study is a refined version from the NVIDIA public model store, which is better than the original ILSVRC VGG model (top-5 error rate is improved from 7.3% to 6.6%)

(1:10) approximates the natural distribution of ROP screening results in our study population. Based on the test result of the three models, the best one was chosen as the final DNN model. 7) Five pediatric ophthalmologists were asked to independently classify the same test set. 8) The performance between DNN and human was compared.

F. STATISTICAL ANALYSIS

The following measures were used: accuracy, precision (PPV, positive predictive value), sensitivity (recall), specificity, F1 score (the harmonic mean of precision and recall), Youden's index (sensitivity + specificity −1), and MCC (Matthews Correlation Coefficient). The result was provided by the ConfusionMatrix module from the pandas_ml (Python Data Analysis Library – Machine Learning) package (v0.5.0). The data from the five pediatric ophthalmologists were plotted in a receiver operating characteristics (ROC) curve and a precision-recall curve (P-R curve) of the DNN to visualize their performances.

III. RESULTS

A. CHARACTERISTICS OF THE INFANTS

The demographics of the premature infants for both data sets are shown in Table 2. There were 45.4% and 8.9% patients who developed ROP in training set and test set, respectively.

B. PERFORMANCE MEASURES

The final layer of the DNNs is a softmax (multinomial logistic regression) layer. It gives probabilities for each class:

$$h_\theta(x_i) = \begin{bmatrix} p(y_i = 0|x_i; \theta \\ p(y_i = 1|x_i; \theta \end{bmatrix} = \frac{1}{\sum_{j=1}^{k} e^{\theta_j^T x_i}} \begin{bmatrix} e^{\theta_0^T x_i} \\ e^{\theta_1^T x_i} \end{bmatrix}$$

The class with the biggest probability is the final prediction. The test set contained 1742 images (155 with ROP and 1587 without). The final test results are shown in Table 3. Based on the highest ROC area under the curve (AUC), we selected VGG-16 as the final DNN model. Nevertheless, the performance of GoogLeNet was very close to that of

TABLE 2. Subject demographics of training and test sets.

Variable	Training Set (N=17801)	Test Set (N=1742)
Gender		
Male	10075 (56.6%)	988 (56.7%)
Female	7726 (43.4%)	754 (43.3%)
Birth Weight (kg)		
Min	0.63	0.78
Max	2.00	2.00
Mean	1.49	1.50
<0.9	374 (2.1%)	33 (1.9%)
0.9-1.1	2866 (16.1%)	289 (16.6%)
1.1-1.3	2937 (16.5%)	286 (16.4%)
1.3-1.5	3667 (20.6%)	359 (20.6%)
1.5-1.8	3970 (22.3%)	390 (22.4%)
1.8-2.0	3987 (22.4%)	385 (22.1%)
Gestational Age (week)		
Min	24	25
Max	36.4	36.2
Mean	31.9	32.0
<27.6	1068 (6.0%)	99 (5.7%)
27.6-29.8	3400 (19.1%)	329 (18.9%)
29.8-32.0	8010 (45.0%)	790 (45.4%)
32.0-34.2	3988 (22.4%)	392 (22.5%)
34.2-36.4	1335 (7.5%)	132 (7.6%)
Diagnosis		
Positive	8090 (45.4%)	155 (8.9%)
Treated cases *	4488	86
Negative	9711 (54.6%)	1587 (91.1%)

* Treated cases: number of infants that were treated for ROP

TABLE 3. Deep neural network performance measures on the test set.

Measurement / Model Type	AlexNet	VGG-16	GoogLeNet
ROC AUC	0.836	0.998	0.997
Accuracy	0.782	0.979	0.980
Sensitivity / recall	0.729	0.987	0.968
Specificity	0.787	0.978	0.982
Precision	0.252	0.818	0.838
F1 score	0.374	0.895	0.898
Youden index	0.516	0.966	0.949
MCC	0.336	0.888	0.89

DNN = deep neural network; ROC = receiver operating characteristics curve; AUC = area under the curve; MCC = Matthews correlation coefficient.

VGG-16. From Table 1 and Table 3, it can also be seen that the classification accuracies of VGG-16 and GoogLeNet on the test set (97.9% and 98.0%) were very close to those on the validation set (97.9% and 98.1%), indicating that the models did not suffer from overfitting and have decent generalization capabilities.

Fig. 3 shows the ROC curves and P-R curves of the three models. VGG-16 and GoogLeNet had much better

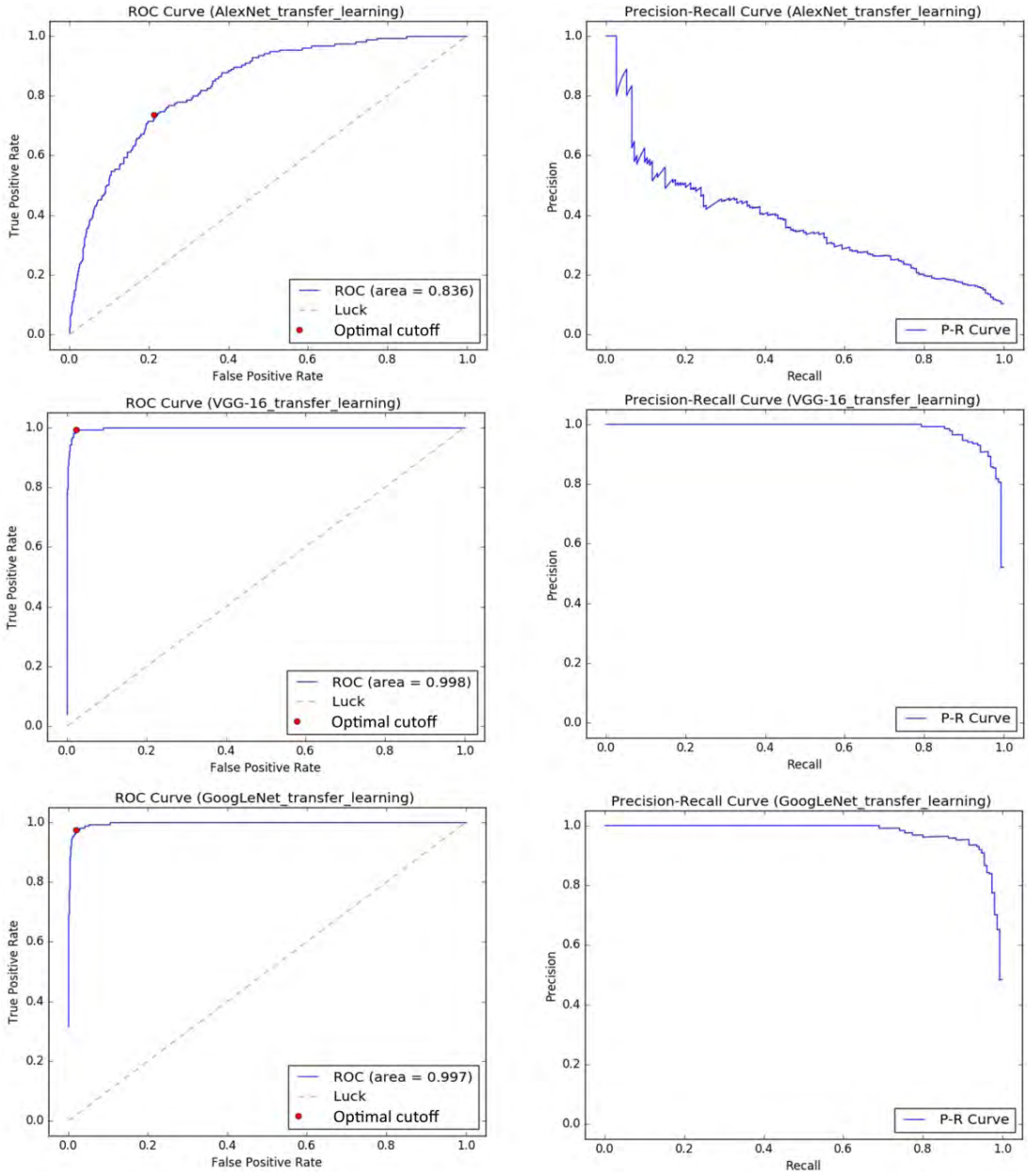


FIGURE 3. ROC curves and precision-recall curves of the candidate DNNs trained by transfer learning. From top to bottom: AlexNet, VGG-16, and GoogLeNet. ROC: receiver operating characteristic; DNN: deep neural network.

performances than AlexNet. Their ROC curves were close to the left-top corner and their P-R curves were close to the right-top corner.

In the screening for ROP, the natural data distribution is unbalanced (ROP:non-ROP = 1:10), so the P-R curve could be more helpful. In most cases, sensitivity or recall (percentage of actual patients who developed ROP that are predicted as positive) is much more important than precision (percentage of predicted positive cases that are actually ROP). In actual clinical settings, we can choose a point in the P-R curve that has a higher sensitivity with a lower yet acceptable precision.

C. PERFORMANCE COMPARISON BETWEEN DNNs AND HUMAN PEDIATRIC OPHTHALMOLOGISTS

Five pediatric ophthalmologists were asked to independently classify the same test data set. The five pediatric ophthalmologists included two senior experts (chief physicians), two attending physicians, and one resident. The results are shown in Table 4. The performance of the five ophthalmologists was also plotted as ROC and P-R curves compared with the DNN (Fig. 4). According to the ROC curve, it can be seen that the performances of both DNN and ophthalmologists were fairly good and quite close. Most ophthalmologists' points lie on or slightly above the ROC curve. In the

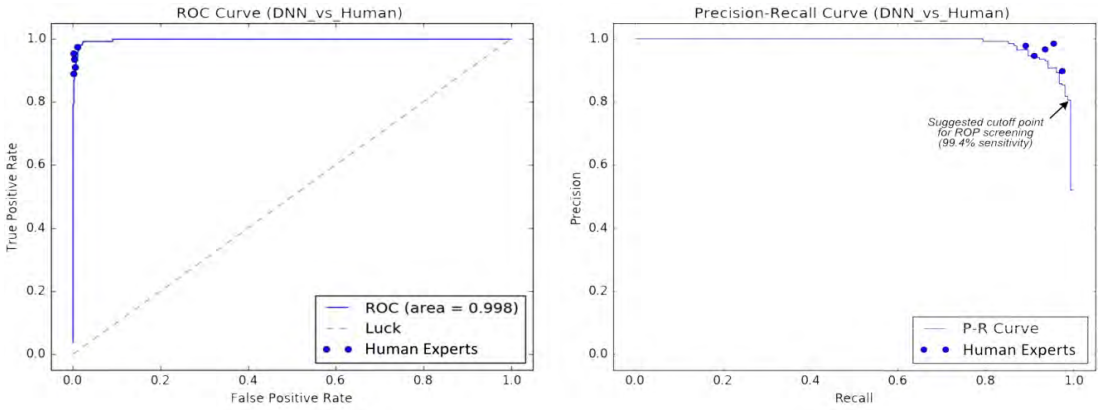


FIGURE 4. The ROC and P-R curves of the ophthalmologists' performance points. The performance of five pediatric ophthalmologists was plotted in the ROC curve and P-R curve of the DNN. ROC: receiver operating characteristic; P-R: precision-recall. DNN: deep neural network.

TABLE 4. Ophthalmologists and deep neural network performances on the test set.

Ophthalmologists and DNN	Accuracy	Sensitivity/ recall	Specificity	Precision	F1 score	Youden index	MCC
Chief 1	99.5%	95.5%	99.8%	98.7%	0.970	0.953	0.968
Chief 2	99.1%	93.5%	93.5%	96.7%	0.951	0.932	0.946
Attending 1	98.8%	89.0%	99.8%	97.9%	0.932	0.888	0.927
Attending 2	98.8%	97.4%	98.9%	89.9%	0.935	0.963	0.929
Resident	98.7%	91.0%	99.5%	94.6%	0.928	0.905	0.921
Median of ophthalmologists	98.8%	93.5%	99.5%	96.7%	0.935	0.932	0.929
DNN cutoff 1	97.8%	99.4%	97.7%	80.6%	0.890	0.970	0.884
DNN cutoff 2	97.9%	98.7%	97.8%	81.8%	0.895	0.966	0.888
DNN cutoff 3	98.8%	96.1%	99.0%	90.9%	0.934	0.951	0.928
DNN cutoff 4	98.8%	94.1%	99.3%	93.0%	0.936	0.935	0.930
DNN cutoff 5	98.8%	93.5%	99.4%	93.5%	0.935	0.929	0.929

MCC = Matthews correlation coefficient; DNN= deep neural network. VGG-16 was selected as the final DNN model.

P-R curve, three ophthalmologists' points lie close to the curve, while two points are above the curve. This means that the DNN model performs closely to three out of five human pediatric ophthalmologists. The two points above the curve correspond to the two chief physicians, which mean their performances are better than the other three non-expert physicians.

For DNN, different cutoff points in the P-R curve correspond to different classification strategies. Different physicians also manifest different classification strategies. For example, some focus on precision to reduce false positive rate, while others concentrate on recall to minimize false negative rate. In order to adapt to such individual differences, we selected five different cutoff points: 99.4%, 98.7%, 96.1%, 94.1%, and 93.5%. The performance measures for each cutoff are shown in Table 4. Cutoff at 94.1% sensitivity maximized the F1 score in the P-R curve. The F1 score is the harmonic mean of precision and recall: F1 score = 2∗precision∗recall / (precision + recall), which is often used for unbalanced data.

The DNN performance at 94.1% sensitivity was 98.8% accuracy, 99.3% specificity, and 93.0% precision. The median ophthalmologist performance was 98.8% accuracy, 93.5% sensitivity, 99.5% specificity, and 96.7% precision. It can be seen that their performances were very close.

As seen from the P-R curve, sensitivity (recall) and precision are two inter-related measures. Increasing one will penalize the other. In actual ROP screening settings, sensitivity (recall) is much more concerned than other measures. The pediatric ophthalmologists suggest using the 99.4% sensitivity cutoff point. Compared to the 94.3% sensitivity cutoff point, precision drops by 12.4% and accuracy drops by 1.0% for a 5.3% increase in sensitivity. This means that the DNN is able to find out more true positive samples but at the cost of allowing more false positive ones.

D. ANALYSIS OF MISCLASSIFIED SAMPLES

After the test, a total of 34 images are misclassified, including 29 false positive (FP) and 5 false negative (FN) samples. Fig. 5 shows several typical false negative (FN) and false

TABLE 5. Performance comparison between fine-tuned models and trained-from-scratch models.

Model Type	Transfer learning		Trained from scratch	
	ROC AUC	Accuracy at best ROC cut-off	ROC AUC	Accuracy at best ROC cut-off
AlexNet	0.836	78.2%	0.819	77.9%
VGG-16	0.998	97.7%	0.894	80.4%
GoogLeNet	0.997	98.0%	0.842	73.9%

ROC = receiver operating characteristics curve; AUC = area under the curve

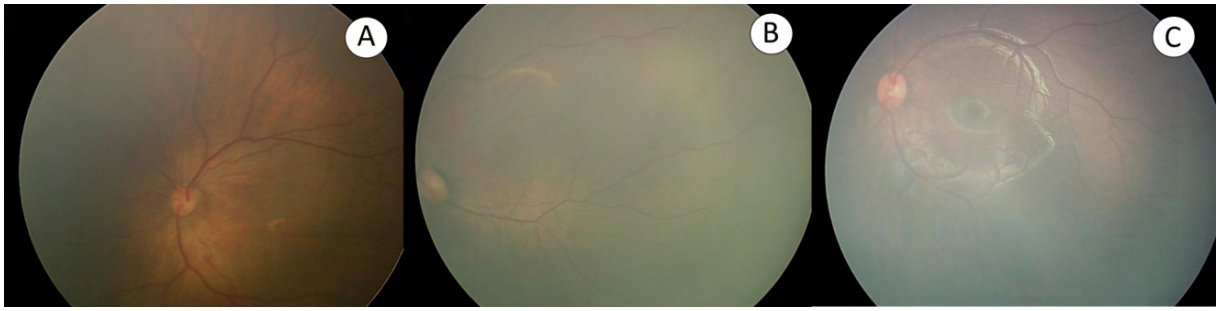


FIGURE 5. Misclassified image samples. (A) A false negative (FN) sample, which shows insignificant characteristics of the plus disease. (B) A false positive (FP) sample with an artifact in the top left region, which was misrecognized as a "ridge" of ROP by the DNN. (C) Another FN sample with more severe artifact which covers the circular area around the macula.

positive (FP) samples (using 94.3% sensitivity cutoff). These misclassified samples were shown to pediatric ophthalmologists to find out the possible causes in order to give a direction for further DNN improvement.

Among the 5 FN samples, 1 sample is misclassified due to vagueness in the marginal area. When the focus is near the optic disk, the peripheral part of the images will be vague due to the camera's focus (the fundus is a curved surface) and lightening (center of the sight is the brightest) designs. The other 4 cases were all related to the plus disease, which is measured by the tortuosity of blood vessels. The judgment of the plus disease does not lie on the two ends of 0 or 1, but usually lies somewhere in between. For instance, the FN sample in Fig. 5 shows insignificant characteristics of plus disease, and pediatric ophthalmologists usually have different opinions for such threshold samples. For such samples, the DNN model's softmax layer outputs probabilities near the classification threshold. Feasible solutions for such FN samples include: 1) providing more typical training data of plus and non-plus diseases; and 2) integrating traditional retinal blood vessel segmentation algorithms and morphological measures to give a more accurate plus score.

Among the 29 FP samples, 18 (62%) were due to vagueness in marginal areas, 3 (10%) were due to excessive light exposure, 8 (28%) were due to optical artifacts. Picture vagueness is a major cause of misclassification, especially for the early stage of ROP, the demarcation line and the ridge structure can be quite difficult to identify, even for human eyes. Development of new-generation fundus imaging instrument and improvement of technicians' photographing skills (e.g. setting proper photograph parameters and getting infants cooperation) will be a possible solution. Optical artifact is another cause for misclassification. These artifacts were not real physiological or pathological structures on the fundus, but were fake textures generated by the lens of the digital camera. In the two FP samples shown in Fig. 5, the DNN model recognized the artifacts as "ridges" of ROP. A possible solution would be providing more data in both classes with artifacts. Although an artifact and a real ridge may appear similar, they have different local visual features. A real ridge usually has neovascularization near it while an artifact does not alter the natural blood vessels. More training data would be helpful for the DNN model to distinguish these subtle local features.

IV. DISCUSSION

Screening for ROP using wide-angle retinal images requires professional knowledge and rich experience. These personal experiences are usually difficult to be completely represented in formal rules or explicit logics. Therefore, it is always a challenge to transfer the valuable experience from senior pediatric ophthalmologists to junior ones. Telemedicine offers an opportunity to provide expertise in underserved areas [24], [25]. The Telemed-R system was designed to support cross-regional telemedicine for ROP screening. The aim of the present study was to evaluate the performance of DNNs against that of pediatric ophthalmologists. Three DNNs were compared and VGG-16 achieved the best performance. Using a sensitivity cutoff of 94.3%, VGG-16 and human performances were comparable. This result is also supported by a recent study by Brown *et al.* [22], who trained a DNN with 5511 retinal

photographs. On their test set, the DNN outperformed six out of eight experts.

DNN-based CAD can be a promising technology that frees pediatric ophthalmologists from the tedious and time-consuming image reading task in screening for ROP. The second advantage of DNN is stability since all images are analyzed in the same way and no subjectivity is involved. Although the gold standard comes from ophthalmologists, humans cannot always perform at their best status, due to fatigue, emotions, and various subjective reasons. DNNs, on the other hand, once properly trained, are free of such errors. The third advantage of DNN is configurability. Based on different task requirements, the DNN can provide corresponding classification strategies, e.g. high sensitivity configuration for ROP screening.

The results in the study verified the feasibility of the designed transfer learning framework. The high-level features and semantics pre-trained on the general-purposed ImageNet data set can be ''transferred'' to the wide-angle retinal image classification task in ROP screening. In addition, to evaluate exactly how much the DNN training could benefit from transfer learning, we also trained three ''from scratch'' DNN models without pre-trained weights. These models were validated on the same test set and the results are shown in Table 5. It can be seen that transfer learning enables significant performance improvement for VGG-16 and GoogLeNet. Their accuracies are improved by 17.3% and 24.1% respectively.

Besides classification, object detection is another deep learning technology that can assist the screening for ROP. In a typical wide-angle retinal image, there are several key physiologic structures such as the optic disk, macula, and fovea. These structures are the landmarks to define the ''zones of ROP'', which are important for the assessment of ROP. There are also ROP-related pathological structures such as ridges, bleeding, laser photocoagulation marks, etc. The detection of such structures can further improve the accuracy and quality of CAD. This object detection research requires pediatric ophthalmologists to annotate corresponding regions in the wide-angle retinal images. The image annotation function is currently under development in Label-R. The detection of such structures is still a limitation of other CAD DNNs for ROP.

Besides screening for ROP, we are planning to integrate the DNN model in an online education system (INTERNS-R), designed for students and interns. The system includes all labeled images as a basic education resource. It also allows users to upload new wide-angle retinal images. Each uploaded image will be pre-labeled by the DNN classifier, which can be further validated by users. In this way, INTERNS-R can provide extra training data for the classifier. It also provides a crowd-souring approach to continuously improve the labeling qualities.

Forward propagation of neural network is executed when making predictions. As DNN is a large scale neural network, a parallel processor (e.g. graphics processing unit (GPU)) is desirable. Unfortunately, in many production situations, especially real-time embedded systems, GPU is not available. To address this problem, there have been efforts to reduce network size and memory usage, e.g. SqueezeNet and DenseNet. In the future, in order to apply the DNN to handheld retina camera devices, we need to further investigate how to simplify the network architecture to boost the propagation speed while maintaining an acceptable accuracy level.

This study has some limitations. Explainability has long been an issue for the neural network family. In most cases, DNN works like a black box, and lacks explainability, as observed in the present study. If users cannot understand how the model works internally, they may not easily accept and trust the model by the end users and regulators. In the future, new techniques, such as hidden layer visualization, will be explored to make DNN more explainable, transparent, and provable. For the routine screening of ROP, the classification of ''disease'', ''not disease'', and ''unqualified'' can well serve routine screening purpose. Nevertheless, the actual ROP-related diagnoses are more complex and fine-grained and include acute ROP, AP-ROP, pre-threshold ROP, regression of ROP, as well as stages, zones, extent and plus diseases of ROP [2], [3], [27]. Many pediatric ophthalmologists require the detection of ''treatment-requiring ROP (an indication for surgical intervention)'' as an urgent next-phase task. To better suit such potential tasks, the current version of Label-R provides a hierarchical tree with all ROP-related diagnosis terms. This diagnosis tree was based on ROP_MST (ROP Minimal Standard Terminology) [28]. In the next phase, we will ask pediatric ophthalmologists to label the wide-angle retinal images with more detailed labels, and train a more fine-grained classifier.

In conclusion, CAD with DNNs shows a comparable performance with human pediatric ophthalmologists in the classification task of wide-angle retinal images for ROP screening. The DNN is a promising technology that can alleviate pediatric ophthalmologists from the routine image reading tasks in screening for ROP. In the future, this DNN can be further integrated into the next-generation portable fundus imaging devices. Such solution will greatly enhance the screening efficiency and accuracy of ROP, especially for under-developed countries and regions where human specialists are insufficient.

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ETHICS APPROVAL

The wide-angle retinal images used in this study were extracted from Telemed-R without other sensitive data. As this study did not involve patients' sensitive data or clinical experiments and since the data set were only confined within the research group, the RCG institutional review board decided that approval and registration were not required for this study.

AVAILABILITY OF DATA AND MATERIALS

Supplemental material has been published under the CC license (DOI: 10.21227/q5jw-t682), including: 1) training and test datasets; 2) model definitions, weights, and graphs of the trained DNNs; 3) performance measures and softmax probabilities of DNNs on the test dataset; 4) performance data of ophthalmologists; 5) scripts used in this study for batch inference and data analysis; 6) response letter to reviews.

COMPETING INTERESTS

All authors declare that they have no competing interests.

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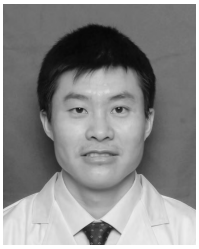
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